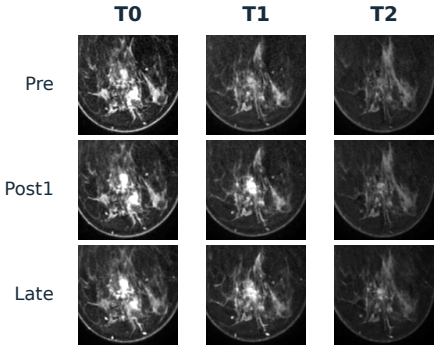
A Observed MRI and continuous visit encoding



Use only examinations available at landmark k

3D convolution shared by phase fine-tuned

Phase concatenation 2 × 4 × 4 pooling 32 spatial tokens

Spatial Transformer 8 layers; d = 768 online encoder E_θ

Continuous features no quantizer or decoder

$X_t \in \mathbb{R}^{3 \times 32 \times 128 \times 128}$

$Z_t \in \mathbb{R}^{32 \times 768}$

B Multimodal fusion, autonomous forecasting, and training supervision

Observed visit tokens clinical values + masks assigned-arm scenario

Fusion Transformer 8 layers 8 queries per transition 4 pCR queries

Autonomous latent predictor $S_k \to \hat{S}_{k+1} \to \cdots \to \hat{S}_3$ 8-layer time-causal network

Terminal state $\hat{S}_3$ change $\hat{S}_3 - S_k$

Current state

Observed MRI

Future MRI: training only

EMA visit encoder $E_{\bar{\theta}}$ $\beta = 0.999$; stop-gradient

Future targets

Patient-equal masked state L1 $\mathcal{L} = \mathcal{L}_{\text{BCE}} + 0.5\mathcal{L}_{\text{roll}}$

Online weights update EMA after each step

C Mean forecast readout

Uniform spatial averaging over N = 32 tokens

$v_k = \frac{1}{N}\sum_n \hat{S}_{3,n}$

$d_k = \frac{1}{N}\sum_n (\hat{S}_{3,n} - S_{k,n})$

Direct q_k

$[q_k; v_k; d_k]$ concatenation

LN + linear sigmoid

p_k

Terminal state and change use uniform token weights.

D Matched ablation design

Model	λ_{roll}	Readout	Pool
NoJEPA	0	Direct	—
DirectReadout	0.5	Direct	—
ForecastMean	0.5	Forecast	Mean

Three matched principal variants